

Context Is the New Weight: Trading the Attention Sink for a Constant Working Memory via Windowed Finetuning

Anonymous submission

Abstract

Causal language models develop an *attention sink*: heads route a large fraction of their attention onto the first one or two tokens, regardless of content. We argue this sink is not an accident but a direct consequence of the triangular causal mask — early positions are visible to every later query, so they accumulate attention — and that it is precisely what makes a pretrained model fragile to a bounded context: naively restricting attention to a sliding window deletes position 0 and the model collapses. We instead finetune the attention itself under a sliding-window mask, so the model operates with a *constant working memory* of the W most-recent tokens and must offload everything older into its weights — *context becomes weight*. We study three variants: a plain **windowed** model, a **startup** model with a trainable cold-start prompt, and a **persistent-prompt** model whose prompts are always attendable (trainable attention sinks). On Qwen2.5-0.5B, windowed finetuning recovers near-full-context language-modelling quality at a 256-token window and drives the position-1 sink from 57% of heads to 0%; the sink relocates onto the startup prompt. On the hybrid Qwen3.5-9B (only 1/4 of layers are softmax attention), finetuning just those 8 layers under a window matches the model’s full-context perplexity at a 256-token window and *beats StreamingLLM without keeping any sink tokens*. A train-window $\times$ inference-window study yields a simple deployment rule — *train small, deploy anywhere* — and the persistent-prompt variant turns the sink into 64 trainable always-on registers that absorb a controllable 19% of attention.

1 Introduction

Transformer language models are trained with a causal (triangular) attention mask: a query at position t may attend only to keys at positions $\leq t$. A robust empirical phenomenon emerges under this mask — the *attention sink* (Xiao et al. 2024): across heads and layers, a disproportionate share of attention mass lands on the very first token(s) of the sequence, even when those tokens carry no relevant information. The sink is now understood as a “no-op” that heads use to dump attention they do not need, and exploiting it (keeping a few initial tokens) is what lets streaming inference work at all (Xiao et al. 2024; Han et al. 2024).

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We take the sink seriously as a *cause*, not just a curiosity. Two observations motivate this paper. First, the sink is an artifact of the mask geometry: under a triangular mask the first token is in the receptive field of *every* later query, so it is structurally positioned to accumulate attention; the data, by contrast, dilute the first token’s importance as context grows. Second, the sink is exactly what breaks a model when we try to bound its memory. If we want a *constant working memory* — attention restricted to the W most-recent tokens, so that compute and KV-cache are $O(W)$ rather than $O(t)$ — then a plain sliding window deletes position 0, removes the sink, and the pretrained model collapses.

The standard response is to keep the sink at inference time (StreamingLLM: retain a few initial “sink” tokens plus the recent window) (Xiao et al. 2024). We ask a different question: *can we finetune the sink away?* If the model is trained under the sliding-window mask, every token must learn to predict from a bounded window, and any long-range information it still needs has to be written into the *weights* rather than read from far-away keys. This is the sense in which “context is the new weight”: adaptation that an in-context model performs by attending to distant tokens is, after windowed finetuning, performed by the parameters.

We make the following contributions:

- We **disentangle two types of attention sink by structure and cause** — a *concentrated* sink on the first token and a *sparse* sink on low-information separators (commas, periods) — orthogonal to the *functional* split (no-op vs. broadcast) of Fesser et al. (2026), and trace them to two *different mechanisms*: the concentrated sink to the triangular mask’s exposure imbalance, the no-op sink to content-driven bounded-memory token usage (Sec. 2–3). We show by finetuning that the positional sink is *learned and removable* — a windowed mask drives it to zero — while the no-op sink remains and is leaned on more.
- We introduce three windowed-finetuning variants — **windowed**, **startup** (a trainable cold-start prompt), and **persistent-prompt** (always-on trainable registers) — that give a model a constant working memory (Sec. 4).
- On Qwen2.5-0.5B and the hybrid Qwen3.5-9B we show that windowed finetuning recovers near-full-context perplexity at a 256-token window, beats StreamingLLM *without* sink tokens, and obeys a clean “train small, deploy

<i>structure</i> (ours)	<i>function</i> (Fesser et al. 2026)	
	no-op	broadcast
Type-1 (concentrated)	null-route anchor	global register
Type-2 (sparse)	delimiter scratch	span summary

Table 1: Two *orthogonal* categorizations of attention sinks. The *functional* axis of Fesser et al. (2026) asks what a sink computes — a *no-op* that suppresses its update by routing to a null token (negligible value norm) vs. a *broadcast* that aggregates and redistributes global information (low-rank output). Our *structural* axis asks the sink’s spatial pattern — a *concentrated* sink on the first token (Type-1) vs. a *sparse* sink spread over low-information separators (Type-2). The axes cross-cut (each cell is realizable): a sink of either structure can serve either function, so the functional split alone cannot assign a *cause*. The structural split can — Type-1 is a mask artifact (H2), Type-2 is content-driven — which is what lets us remove one while keeping the other.

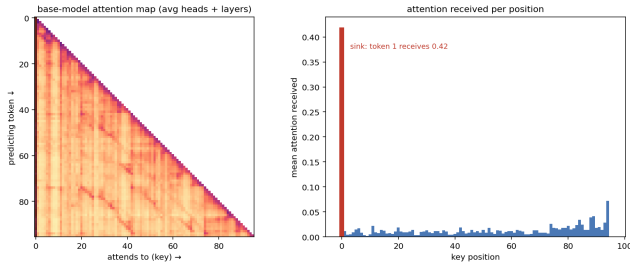


Figure 1: Type 1 (positional) sink in a base causal LM. *Left*: attention map (averaged over heads and layers); the first column is a bright vertical stripe. *Right*: mean attention received per key position — token 1 absorbs 0.42, every other token < 0.05 .

anywhere” rule (Sec. 5).

2 Categorizing Attention Sinks

Attention sinks admit more than one categorization. We propose a *structural* one — by the spatial pattern of the sink itself — distinguishing a *concentrated* sink from a *sparse* sink. They differ in *where* the attention lands and in *what* drives it, and the rest of the paper turns on telling them apart.

Type 1 — the concentrated (first-token) sink. A large, almost content-independent share of every query’s attention *concentrates* on the first one or two tokens of the sequence, regardless of their content. Figure 1 shows this for a base model: the leftmost column of the attention map is a bright vertical stripe, and token 1 absorbs on average 0.42 of the attention mass — an order of magnitude more than any later key — across a majority of heads. This is the sink exploited by StreamingLLM (Xiao et al. 2024): it is tied to absolute *position*, not to meaning.

Type 2 — the sparse (separator) sink. A second sink is *sparse* rather than concentrated: it spreads across semantically near-empty separator tokens *within* the sequence —

punctuation such as commas and periods, delimiters, and other filler. These tokens carry little predictive information of their own; a head with nothing to retrieve at a given step dumps its attention onto the nearest such token, using it as a local “no-op” or scratch register. Unlike Type 1, the no-op sink is distributed throughout the text and keyed to token *identity* rather than to absolute position.

Functional vs. structural categorization. Prior work separates sinks by *function*, but not by *structure* or by *cause*. Fesser et al. (2026) distinguish two algorithms behind a sink — an adaptive *no-op* that suppresses the update by routing to a null token (negligible value norm) and a *broadcast* that aggregates and redistributes global information (low-rank output) — while earlier accounts simply retain the initial tokens at inference (Xiao et al. 2024; Han et al. 2024). We categorize *orthogonally and structurally*, by the sink’s spatial pattern: a *concentrated* first-token sink (Type 1) vs. a *sparse* separator sink (Type 2). The two axes cross-cut (Table 1) — a sink of either structure can serve either function — so the functional split alone cannot assign a *cause*. The structural split can, and this is our central claim (H1): the two structural types arise from *two different mechanisms*. Type 1 is an artifact of the triangular mask’s positional exposure imbalance (H2): a property of the *architecture* that disappears when the mask changes. Type 2 is content-driven, keyed to token *identity*: a property of the *data and the model’s use of it* that persists — indeed is leaned on *more* — once Type 1 is removed (H3). Separating the sink *by position* is what lets us remove the harmful, mask-induced one while keeping (and exploiting) the useful one. We state these mechanisms as testable hypotheses next.

3 Hypotheses

H1: the attention sink is two disentangleable types with distinct causes. We hypothesize that what is usually called “the attention sink” is in fact two phenomena that can be separated. *Type 1* is *concentrated* — a large, content-independent share of attention parked on position 0. *Type 2* is *sparse* — attention spread across semantically near-empty separator tokens (commas, periods, delimiters) used as local no-op registers, keyed to token *identity* rather than to position. Prior work separates sinks by *function* (Fesser et al. 2026) but not by structure or cause; we claim the two *structural* types arise from *two different mechanisms* (H2 and the data, respectively) and can therefore be removed or kept independently.

H2: the Type 1 sink is caused by the context-length imbalance of the causal mask. Under a length- C lower-triangular mask a query early in the sequence has almost no tokens to attend, while position 0 sits in the receptive field of *every* later query — the maximal, content-free exposure. We hypothesize that this imbalance — early queries starved of context, position 0 always available — biases training to recruit the earliest positions as a stable anchor, producing the Type 1 sink. Consistent with this, the measured sink *grows* with context (Fig. 2): the model sinks *against* the data-side first-token importance ($\propto 1/C$, which *shrinks* with context) — the signature of an active, mask-induced mechanism rather than a property of the data.

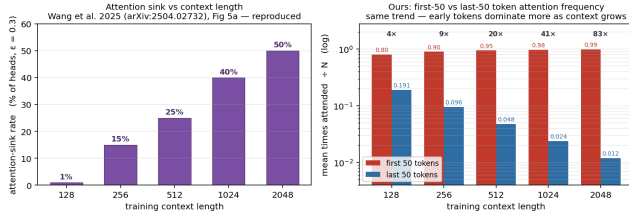


Figure 2: The measured Type 1 sink grows with context length, the opposite of the data-side first-token importance ($\propto 1/C$) — the model sinks *against* the data, an artifact of the mask (H2).

H3: removing the Type 1 sink makes later tokens resort to the Type 2 separators. If the Type 1 anchor is removed — e.g. by a windowed objective that takes position 0 out of every deep query’s receptive field — the storage/no-op function is not lost. We hypothesize that, rather than anchoring the shared representation on the beginning of the sequence, later tokens *resort* to the Type 2 low-information separators, repurposing them to carry the role position 0 used to play and redistributing attention onto them.

H4: separator sinks form a hierarchical store that broadcasts better in streaming. Because Type 2 tokens are distributed through the text — each one sitting at the end of a span of already-accumulated context — we hypothesize that attention parked on them *absorbs* that preceding-token information, building a *hierarchical* representation (separators at clause-, sentence-, and paragraph-scope summarizing progressively larger spans) with more granular control over information broadcasting than a single distant position-0 anchor. In a bounded-window deployment a recent separator is always inside the window and has already integrated its local context, making it a better relay point than a position-0 sink the window has scrolled past — so a constant-working-memory model should not merely tolerate the loss of the Type 1 sink but *benefit* from leaning on Type 2 relays, improving streaming performance.

Why this matters for bounded memory. A constant working memory restricts attention to a window of the W most-recent tokens. For a query deep in the sequence this window no longer contains position 0, so the Type 1 sink is gone; a pretrained model that relies on it produces degenerate distributions and its perplexity explodes. StreamingLLM (Xiao et al. 2024) patches this at inference by always retaining a few initial tokens. We instead remove the dependency by training (Section 4), and ask whether the model falls back on the Type 2 no-op sinks (H3) and whether doing so improves streaming behaviour (H4).

4 Method: Windowed Attention Finetuning

Let the causal mask be $M_{\text{causal}}(t, k) = 1$ iff $k \leq t$. We replace it during finetuning with a sliding-window mask

$$M_{\text{win}}(t, k) = \mathbb{1}[t - W < k \leq t],$$

so each token attends only to the W most-recent positions. We finetune (continued pretraining on generic text, language-

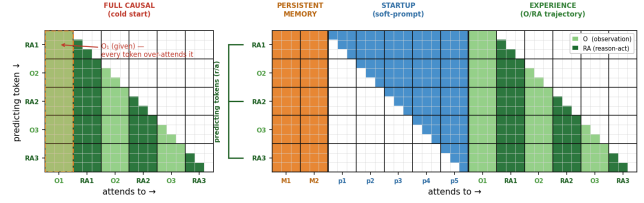


Figure 3: Windowed attention finetuning. The triangular mask (left) is replaced by a sliding window (a band of width W), giving each token a constant working memory of the recent W tokens; older context must be carried by the weights.

modelling loss only — no task supervision) so that the comparison isolates the effect of the *mask*, not of any downstream data. We study three schemes (Fig. 3):

Windowed. The plain sliding window above. The window’s cold start (the first $W-1$ positions, whose window is not yet full) is filled with real preceding tokens during training.

Startup. We prepend P trainable soft prompts that fill the window’s cold start in place of real history, then window over the prompt+content sequence. The prompt slides out of the window for deep tokens, so it functions purely as a learned cold-start anchor — a trainable replacement for the sink token. This grants *availability*: the model can be deployed from token 1 without needing W real tokens of history.

Persistent prompt. We prepend P trainable prompts that are *always* attendable by every content token (a persistent learned prefix), while the content is windowed to W . Each token then sees W recent tokens *plus* P constant learned registers (effective memory $W+P$, of which P is fixed). The prompts become trainable, always-on attention sinks.

5 Experiments

We evaluate on Qwen2.5-0.5B (a pure-softmax-attention model, our proof of concept) and Qwen3.5-9B (the target model). Finetuning is continued pretraining on WikiText-103 (Merity et al. 2017); evaluation is held-out perplexity scored within a working-memory window, plus downstream long-context QA on LongBench (Bai et al. 2024). Unless noted, the window for the headline runs is $W=256$. We organize the evidence by hypothesis.

5.1 H1: The Sink Is Two Disentangleable Types

The two sink types are *separable*. On Qwen2.5-0.5B we measure, under each scheme’s deployment, the attention received by the first token (the concentrated Type-1) versus low-information separators (the sparse Type-2), and the broadcasting entropy of the per-position attention (Table 2, Fig. 4). They move *independently*: continued full-causal (triangle) training keeps — even slightly intensifies — Type-1 (0.37 $\rightarrow$ 0.38), while the **startup** variant drives Type-1 to 0.004 and the Type-2 mass *rises* to 0.40. One type can be removed while the other is kept, which is what it means for

	Type-1	Type-2 (no-op)		broadcast
scheme	(pos-0)	punct.	low-surp.	entropy
base	0.368	0.146	0.148	4.51
triangle-CPT	0.383	0.137	0.138	4.43
windowed	0.151	0.323	0.354	5.18
startup	0.004	0.396	0.449	5.52

Table 2: Where attention goes (Qwen2.5-0.5B; all schemes measured identically at a *genuine* 256-token window over 512-token sequences). Fraction of attention received by the position-0 token (Type-1) vs. low-information no-op tokens (Type-2), the latter measured two ways — a fixed punctuation list and the bottom-20% unigram-surprisal tokens — which agree, so the no-op sink is not an artifact of the punctuation proxy. Continued full-causal (triangle) training keeps (slightly intensifies) the Type-1 sink; windowed and startup training drain it onto the distributed Type-2 tokens and raise the broadcasting entropy. The attention-surprisal correlation sharpens in step (-0.07 at base $\rightarrow -0.16$ at startup): as Type-1 dies, attention concentrates more on low-information tokens.

them to be disentangleable. The split is robust to the Type-2 definition: the same shift appears whether Type-2 is a fixed punctuation list or the bottom-20% unigram-surprisal tokens, and the attention-surprisal correlation sharpens from -0.07 to -0.16 — it is not an artifact of the punctuation proxy.

Structural meets functional. The two structural types also differ *functionally* in the sense of Fesser et al. (2026): across all training schemes the concentrated Type-1 sink carries a low relative value norm (~ 0.4 , the no-op signature — it is read precisely to discard the update), while the sparse Type-2 separators carry near-normal value (~ 0.8 – 0.9) and broadcast content (Table 3). As windowed and startup training drain Type-1, attention shifts onto the higher-value Type-2 tokens — structurally concentrated $\rightarrow$ sparse and functionally no-op $\rightarrow$ broadcast in step. Two caveats keep this honest: pos-0 is not a *pure* no-op (heavily attended, it still injects $\sim 15\%$ of the base model’s output energy), and our *uniform-window* scheme reduces Type-1 *without* the Type-2 spread — it sinks on each window’s relative boundary instead.

5.2 H2: The Concentrated Sink Is a Mask Artifact

Learned and removable. Measuring the position-1 sink before and after finetuning shows it is a learned behaviour. On Qwen2.5-0.5B, a majority of softmax heads ($\approx 57\%$) sink onto token 1 in the base and full-attention-finetuned models; windowed finetuning drives this to 0% (Table 4), and the **startup** variant relocates the sink onto the trainable prompt rather than the content. On the larger hybrid model the absolute sink is weaker but the *direction* is identical: full-attention finetuning *builds* the sink, windowed finetuning *shrinks* it (Fig. 5).

Windowed finetuning recovers full-context quality. Scoring held-out perplexity as a function of the deployed window shows the expected collapse for a base model win-

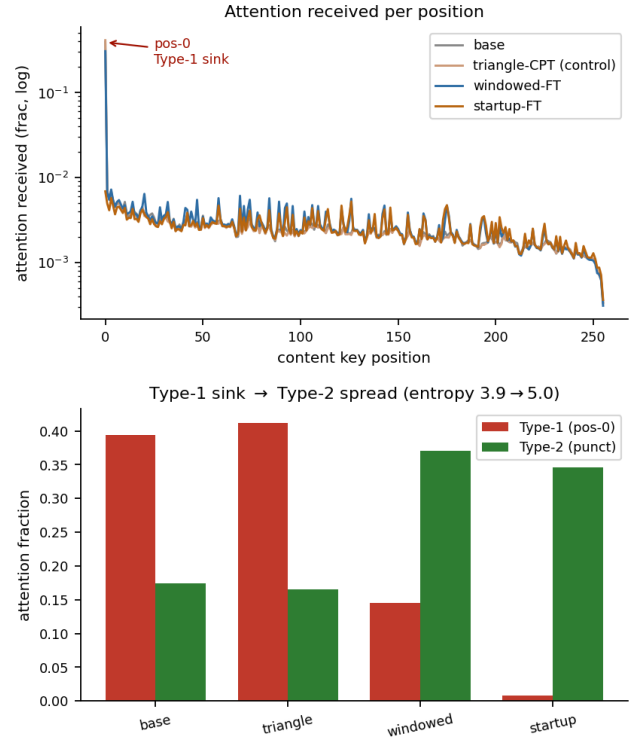


Figure 4: Where attention goes (Qwen2.5-0.5B, all schemes at a genuine 256-token window). **Top:** continued full-causal (triangle) training keeps the Type-1 pos-0 sink; windowed and startup training drain it and the mass moves onto distributed Type-2 no-op tokens — identically whether Type-2 is a punctuation list or the bottom-20% unigram-surprisal tokens (the two bars agree), so it is not a punctuation artifact. **Bottom:** broadcasting entropy rises and the attention-surprisal correlation sharpens ($-0.07 \rightarrow -0.16$) as the positional sink dies.

dowed at inference, and a near-flat curve for the windowed-finetuned model (Fig. 6): a model trained for a 256-token window matches the base model’s full-context quality while attending to $8\times$ fewer tokens (Table 5). Continued pretraining under *full* attention does not help — it amplifies sink reliance — whereas the windowed objective removes it.

Scaling to a hybrid model. Qwen3.5-9B is a hybrid: of its 32 layers, only 8 (every 4th) are softmax attention; the rest are linear/Mamba-style attention (Gu and Dao 2024; Yang, Kautz, and Hatamizadeh 2024). The sink, and the windowing question, therefore concern only those 8 layers — $3/4$ of the network is already a constant-memory mechanism. We full-finetune just the 8 softmax blocks (freezing the 24 linear layers; 8-bit AdamW with gradient checkpointing on a single 48 GB GPU). Even on this model, bounding the softmax window to 256 tokens collapses the base from a full-context perplexity of 6.5 to 20.0; windowed finetuning restores it to 6.3 — full-context quality at a 256-token softmax window — and, unlike the baselines, gains nothing from StreamingLLM sink tokens because it no longer depends on them (Table 6).

scheme	structural (share)		functional ($\langle \ v\ /\text{mean} \rangle$)	
	Type-1	Type-2	at T-1	at T-2
base	0.37	0.08	0.38	0.92
triangle-CPT	0.38	0.07	0.41	0.89
windowed	0.16	0.25	0.43	0.80
uniform-window	0.21	0.08	0.45	0.91
startup	0.004	0.32	0.88	0.81
persist	0.005	0.10	0.86	0.89

Table 3: Structural $\times$ functional transition across training schemes (Qwen2.5-0.5B, streaming eval sliding a $W=256$ window through the sequence). *Structural*: attention share on the concentrated first token (Type-1) vs. the sparse separators (Type-2). *Functional*: relative value norm at each sink ($\|v\|/\text{mean}$; low $\Rightarrow$ no-op, after Fesser et al. 2026). The concentrated Type-1 sink carries a *low* value norm (~ 0.4 , no-op-like); the sparse Type-2 separators a near-normal one (~ 0.8 – 0.9 , value-carrying / broadcast). As windowed and startup training drain Type-1, attention moves onto the higher-value Type-2 — concentrated $\rightarrow$ sparse and no-op $\rightarrow$ broadcast in lockstep. (Caveats: the pos-0 sink is heavily attended, so despite its low value it still injects $\sim 15\%$ of base-model output energy — not a *pure* no-op; and uniform-window reduces Type-1 without the Type-2 spread, sinking on each window’s relative boundary instead.)

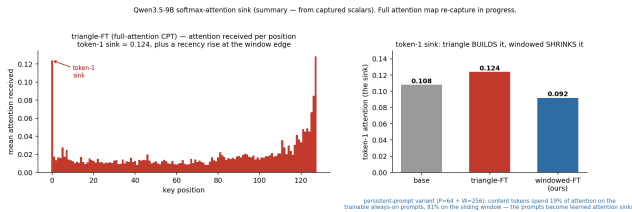


Figure 5: Qwen3.5-9B softmax-layer sink. Full-attention (triangle) finetuning builds the token-1 sink to 0.124; windowed finetuning shrinks it to 0.092; base is between (0.108). The persistent-prompt variant relocates $\sim 19\%$ of every content token’s attention onto trainable always-on registers.

Train small, deploy anywhere. Evaluating every windowed model at every inference window (Table 6) reveals an asymmetry. A model trained at a *small* window (windowed-128) is nearly flat across *all* deployment windows (7.44 $\rightarrow$ 6.62): having learned to predict from 128 tokens, it is robust to any budget. A model trained at a *wide* window is fragile when deployed narrower (windowed-1024 rises to 24.7 at deploy-128), because it learned to lean on context that is no longer present. A model is always fine deployed *wider* than it was trained, and breaks deployed *narrower*: the practical rule is to train at the smallest window one might deploy.

A strict softmax-vs-gated control. Is the gain from *windowing*, or merely from more finetuning? We answer with a strict control: two 1.3B models from the flash-linear-attention suite trained on the *same* 100B tokens with the *same* skeleton, differing only in the token-mixer — a softmax Transformer

	full attention (learned tendency)	windowed mask (as deployed)
base + history	57.6%	0.0%
sliding-history (trained)	5.8%	0.0%

Table 4: Position-1 sink on Qwen2.5-0.5B given the *same* real history (63 history + 256 content, $W=64$): % of softmax heads with mean attention > 0.3 to the history’s first token. Windowed finetuning cuts the learned sink $\sim 10\times$ (57.6% $\rightarrow$ 5.8%); the window masks it to 0% at deployment.

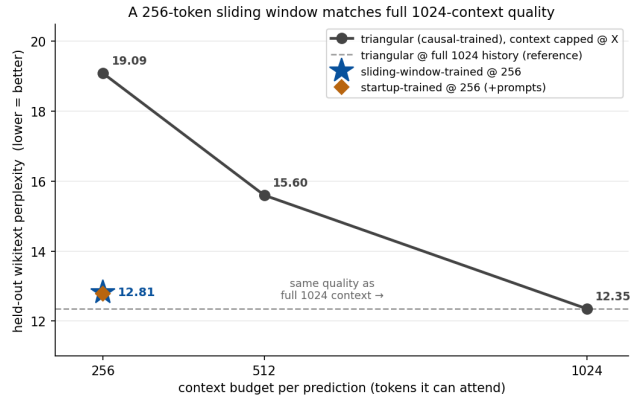


Figure 6: Held-out perplexity vs. deployed working-memory window (Qwen2.5-0.5B). The base model collapses as the window shrinks; the windowed-finetuned model is nearly flat and matches full-context quality at a small window.

and a gated-linear (GLA) model — evaluated across the full train-window $\times$ deploy-window grid (Table 7). The softmax base collapses at tight windows (10.2 $\rightarrow$ 17.9 at deploy-128); continued *full-causal* pretraining (triangle-CPT) only partly recovers (14.0, the control held fixed); windowed finetuning gives near-flat rows — the same *train small, deploy anywhere* asymmetry as the hybrid model — so the recovery is from the window, not the extra training. The gated GLA has no attention window: in its native $O(1)$ -state mode it reaches 8.71 with constant memory, beating every windowed-softmax cell — but only because it always uses full context. A pure recurrence has no honest hard- W window — the only way to impose one is to reset its state, which no real deployment does — so the legitimate comparison is bounded- $O(W)$ sliding-window softmax against native- $O(1)$ gating. There the windowed-softmax retrofit reaches the gated model’s quality at a moderate budget, while the gated model needs no windowing at all.

Compute. The payoff is efficiency. Full softmax attention is $O(L^2)$; the windowed and gated mechanisms are $O(L)$. At a 32k context the windowed model runs $2.7\times$ faster than full attention (Table 8), with linear rather than quadratic growth.

model (Qwen2.5-0.5B)	context budget	held-out ppl
triangular (causal-trained)	256	19.09
triangular (causal-trained)	512	15.60
triangular (causal-trained)	1024 (full)	12.35
sliding-window-trained	256	12.81
startup-trained	256 (+ prompts)	12.78

Table 5: Held-out wikitext perplexity vs. context budget on Qwen2.5-0.5B. The causal-trained model needs the full 1024-token history (12.35) and degrades steeply when windowed (19.09 at 256 — worse than the untrained base, because windowing masks the first-token sink it leans on); the windowed-finetuned model reaches 12.81 at a 256-token window, matching the causal model’s full-context quality on a quarter of the context.

model	perplexity at inference window				
	128	256	512	1024	2048
<i>untrained baseline</i>					
base 9B	59.8	28.1	13.8	7.80	7.26
+ StreamingLLM	9.33	8.47	7.91	7.58	7.28
<i>continued-pretraining baseline</i>					
triangle-FT	33.7	18.2	8.66	6.96	6.59
+ StreamingLLM	7.88	7.39	7.03	6.83	6.56
<i>ours — windowed (one model per training window)</i>					
windowed-128	7.44	7.14	6.94	6.83	6.62
windowed-256	7.82	7.19	6.93	6.79	6.57
windowed-512	11.5	7.82	6.95	6.79	6.55
windowed-1024	24.7	12.7	7.44	6.83	6.54
windowed-2048	27.6	14.5	7.76	6.84	6.53
<i>ours — startup (windowed + cold-start prompt)</i>					
startup-256	7.82	7.17	6.92	6.79	6.57

Table 6: Qwen3.5-9B held-out perplexity, train-window (rows) $\times$ inference-window (columns); bold is native (train = deploy). Windowed models beat both baselines and their StreamingLLM versions at tight windows, with no sink tokens. Small-window models are robust to any deployment window; wide-window models break when deployed narrower (*train small, deploy anywhere*).

5.3 H3: Removing Type-1, the Model Resorts to the Separators

Removing the concentrated sink does not destroy the attention mass — it *relocates* it. The same measurement (Table 2), read causally, shows that as Type-1 drains across triangle→windowed→startup (0.38 → 0.15 → 0.004) the sparse Type-2 mass *rises* in lockstep (0.14 → 0.32 → 0.40), with the broadcasting entropy climbing 4.4 → 5.5: with the positional anchor suppressed, the model resorts to the content-based no-op separators as its store. We note the boundary, addressed under H4: this broader spread is a *distributional* property, not long-range retrieval.

5.4 H4: Separator Sinks Support Streaming

Persistent prompts as trainable sinks. The persistent-prompt variant ($P=64$ always-on prompts + a $W=256$ win-

model	perplexity at inference budget				
	128	256	512	1024	full
<i>softmax — untrained / control</i>					
base	17.9	14.9	12.8	11.3	10.2
triangle-CPT	14.0	12.1	10.8	9.61	8.80
<i>softmax — ours: sliding-window (one model / train window)</i>					
windowed-128	10.6	9.70	9.30	9.10	9.06
windowed-256	10.8	9.63	9.17	8.96	8.92
windowed-512	11.9	9.85	9.15	8.89	8.84
windowed-1024	12.7	10.5	9.36	8.88	8.81
<i>gated (GLA) — same data & skeleton</i>					
gla-CPT	8.71 (window-invariant, $O(1)$ memory)				

Table 7: Strict control on the flash-linear-attention 1.3B pair (same GPT skeleton, same 100B tokens; only the token-mixer differs — softmax vs. gated-linear), held-out perplexity, training budget (rows) $\times$ inference budget (columns); bold marks each softmax model’s native window. **Softmax:** the base collapses at tight windows; the full-mask (triangle) control only partly recovers; sliding-window finetuning gives near-flat rows — *train small, deploy anywhere*. **Gated:** GLA continued-pretrained on the same data (gla-CPT) is *window-invariant* (8.71 at $O(1)$ memory): it has no attention window to tighten, so it runs at a single operating point. The honest comparison is therefore bounded- $O(W)$ sliding-window softmax vs. native- $O(1)$ gating — the windowed-softmax retrofit reaches the gated model’s quality at a moderate budget, while the gated model needs no windowing at all. (A pure recurrence cannot be given a hard W -window without resetting its state, which no real deployment does; we therefore do not report a “windowed gla”.)

dow) lets content tokens spend a controllable fraction of attention on the learned registers: in our run, 19% of every content token’s attention goes to the 64 persistent prompts and 81% to the window. The prompts thus become explicit, trainable attention sinks — the function the first token was being abused for, now a designed component. On perplexity the persistent and startup variants track the plain windowed model, with a small edge when deployed *narrower* than trained (the prompt fills the coldest part of the window). The registers matter most on a downstream task that needs distant context, which we test with a *fair* control: triangle (full-mask), windowed, and persist models all continued-pretrained on the same wikitext for the same steps, so only the mask differs (Table 9). At a 256-token window the full-mask control collapses (HotpotQA generation F1 0.036), windowed training recovers it (0.095), and the persistent-prompt model reaches 0.166 — matching the base+StreamingLLM post-hoc trick (0.166), but as a trained constant-memory model. This marks the boundary of the broadcasting effect: windowed training spreads attention onto the Type-2 tokens (a distributional change) but does not by itself relay a *specific* distant fact; the persistent registers are what restore long-range anchoring, unifying constant-memory streaming with retrieval in one trained model.

seq. length	full	windowed	gated
8k	313	226	258
16k	797	436	499
32k	2363	866	989

Table 8: Forward latency (ms) vs. sequence length (fla 1.3B, one A6000). Full softmax attention scales quadratically; windowed and gated scale linearly — at 32k the windowed model is $2.7\times$ faster than full.

variant (CPT)	deploy	F1	ans-ppl
triangle (full-mask)	windowed	0.036	112
windowed	windowed	0.095	32
persist	persist	0.166	21
base (no CPT)	full	0.305	2.8
base + StreamingLLM	streaming	0.166	13.6

Table 9: HotpotQA, *fair* mask-only control (Qwen2.5-0.5B). The three CPT variants see the *same* wikitext for the *same* steps — only the attention mask differs — so QA-ability drift is held constant. At a 256-token window, the full-mask (triangle) control collapses (F1 0.036); windowed training recovers it (0.095); the persistent-prompt variant reaches 0.166, *matching the base+StreamingLLM post-hoc trick* as a trained constant-memory model. SQuAD-normalised generation F1.

Downstream long-context QA. To isolate the mask effect from task learning we also continued-pretrain (no QA supervision) and evaluate zero-shot on LongBench multi-hop QA (Bai et al. 2024). At a tight 256-token window, the windowed model uses its bounded memory markedly better than a full-attention-pretrained control, and StreamingLLM helps the baselines but not the windowed model — the same signature as in perplexity, on a real task (Fig. 7).

6 Related Work

Attention sinks and streaming. Xiao et al. (2024) identified the sink and exploited it for streaming inference by retaining initial tokens; Han et al. (2024) use a similar fixed pattern for length generalization. These keep the sink at inference; we remove the dependency by finetuning, and (in the persistent-prompt variant) replace the accidental sink with trainable registers.

Windowed and efficient attention. Sliding-window attention is used architecturally by Longformer (Beltagy, Peters, and Cohan 2020) and Mistral (Jiang et al. 2023), and linear/state-space models (Gu and Dao 2024; Yang, Kautz, and Hatamizadeh 2024) achieve constant memory by construction; Qwen3.5 is a hybrid of the two. We instead *finetune* a pretrained softmax model into a windowed one and study what the sink does under that change. Position-extrapolation methods such as ALiBi (Press, Smith, and Lewis 2022) share the “train short, test long” spirit; we observe a complementary “train small, deploy anywhere” rule for the attention window.



Figure 7: LongBench multi-hop QA (Qwen2.5-0.5B, continued-pretraining, no task supervision). At a tight window the windowed model uses a bounded working memory better than a full-attention control; the gap shrinks as the window grows.

Context as weight. Our framing — that windowing forces older context into the parameters — connects to context distillation and knowledge editing; here the mechanism is attention geometry rather than an explicit distillation loss.

7 Conclusion

The attention sink is a learned consequence of the triangular causal mask, and it is exactly what makes a pretrained model fragile to a bounded context. Finetuning under a sliding-window mask removes the dependency: the model acquires a constant working memory, matches full-context quality at a small window, and — unlike inference-time fixes — needs no sink tokens. Two design points fall out: “train small, deploy anywhere,” and the persistent-prompt variant that turns the accidental sink into a controllable set of trainable registers. We view this as evidence that, for the long-range part of a model’s behaviour, context can be traded for weight.

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